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Characterization of a BODIPY Dye as an Active Species for Redox Flow Batteries

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An all-organic redox flow battery (RFB) employing a fluorescent boron-dipyrromethene (BODIPY) dye (PM567) was investigated. In a RFB, the stability of the electrolyte in all charged states is critically linked to coulombic efficiency. To evaluate stability, bulk electrolysis and cyclic voltammetry (CV) experiments were performed. Oxidized and reduced, PM567 does not remain intact; however, the products of bulk electrolysis evolve over time to show stable redox behavior, making the dye a precursor for the active species of an RFB. A theoretical cell potential of 2.32 V was predicted from CV experiments with a working discharge voltage of approximately 1.6 V in a static test cell. Mass spectrometry was used to identify the products of bulk electrolysis. Related experiments were carried out using ferrocene and cobaltocenium hexafluorophosphate as redox-stable benchmarks to further explain the stability results. The coulombic efficiency of a model cell using PM567 as a precursor for charge carriers stabilized around 73 %.

Introduction

Redox flow batteries (RFBs) have gained momentum in recent years as large capacity energy storage systems that can be used in tandem with wind and solar power generation to provide a stable supply of renewable energy.[1–3] In an RFB, external reservoirs are utilized to store electrolyte solutions, typically comprising dissolved electroactive species, supporting electrolyte, and solvent. The use of external reservoirs enables an independent scaling of power and energy densities within a system as the former is controlled by electrode surface area and the latter by the total volume of electrolyte solution. Electroactive species stored in external tanks are pumped over the electrode surfaces during charge and discharge processes, resulting in oxidations and reductions serving as the impetus for energy storage or use, respectively.

Since the development of pioneering RFB designs,[4] a variety of electrolyte materials have been evaluated as electroactive species in aqueous and non-aqueous RFBs. Aqueous RFB chemistries include chromium/iron,[5] vanadium chloride/polyhalide,[6] bromine/polysulphide,[7] zinc/bromine,[8] and all-vanadium.[9,10] The generation of hydrogen and oxygen via water splitting ($E^0 = 1.23$ V) limits the working voltages of such systems. In practice, the over potentials associated with O_2 and H_2 evolution widen the solvent window depending on the electrode material used, a subject of great interest in the design of electrolysis systems where hydrogen and oxygen are desired products.[11]

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Non-aqueous RFBs suffer from high internal resistance and lower solubility relative to aqueous designs. These limitations motivate research to maximize conductivity and solubility in organic solvents, identify new redox active agents, and develop new membrane materials optimized for use in non-aqueous environments.[12] Matsuda et al. studied tris(2,2′-bipyridine)-ruthenium(II) tetrafluoroborate as the active species in a non-aqueous RFB that had an open circuit potential of 2.6 V.[13] Chakrabarthi et al. reported a non-aqueous RFB with ruthenium acetylacetonate as the active material showing an open circuit potential of 1.8 V.[14,15] The library of coordination complexes evaluated for RFBs has grown since these initial studies to include vanadium acetylacetonate,[16] chromium acetylacetonate,[17] manganese acetylacetonate,[18] 1,10-phenanthrolinecobalt(II) hexafluorophosphate,[19] and bis(acetylacetone)ethylenediamine cobalt.[20] Organic molecules are also suitable as active species in RFBs, with TEMPO[21] and modified quinones[22,23] among the scaffolds that have been studied towards this end.

These compounds and complexes underscore some important parameters regarding active species in RFBs, including solubility and long-term stability. Decomposition of the charge carriers has a direct impact on energy storage (coulombic efficiency) as well as device lifetime. In some cases, the initial redox-active species may act as a precursor that forms an electrolyte mixture once charged. In other cases, irreversible degradation may occur, requiring electrolyte regeneration.[24] A knowledge of redox behavior and stability serves as a guide to understand the performance of an RFB and aids in the design of next-generation electrolytes. Generally, cyclic voltammetry (CV) studies are employed to evaluate the reversibility and kinetic behavior of active species, along with charge–discharge studies that show long-term cycling stability. A caveat associated with CV is that the timescale of a typical experiment is only seconds to minutes. Therefore, a molecule that appears reversible by CV may exhibit significant redox changes when it remains charged even for relatively short amounts of time (minutes to hours).

Herein, we have evaluated a boron-dipyrromethene (BODIPY) dye (PM567), as a precursor for the active species in both positive and negative electrolyte solutions of an RFB. BODIPY dyes are extensively studied for their interesting electrochemical, photochemical, and electrogenerated chemiluminescence properties.[25,26] It was reported that PM567, a variant in which all core C–H sites are substituted, shows two well-separated and reversible redox couples even at slow scan rates (ca. 50 mV s^{-1}) indicating that the radical ions formed upon oxidation and reduction are stable at the CV timescale.[26,27] The occurrence of two reversible redox events makes PM567 a potential candidate as an active species in an all-organic RFB. Bulk electrolysis, UV/Vis, and mass spectrometry experiments have been used to interrogate the stability of oxidized and reduced PM567 stored over the course of one week. Related studies were performed using ferrocene and cobaltocenium hexafluorophosphate that serve as models for oxidation and reduction stability, respectively. A model cell was used as the basis for charge–discharge experiments indicating that despite the lack of long-term stability of PM567, the decomposition products exhibit redox stability, giving rise to a plateauing coulombic efficiency over the course of 100 cycles.

Results and Discussion

Voltammetric behavior of PM567

The CV of 6 mM PM567 dissolved in 0.1 M tetrabutylammonium hexafluorophosphate ($TBAPF_6$) in anhydrous acetonitrile at a scan rate of 0.5 V s^{-1} is shown in Figure 1. For comparison, a CV of 0.1 M $TBAPF_6$ in anhydrous acetonitrile is also shown with no significant current response in the region of interest (−2–1 V vs. Ag/Ag$^+$). The CV profile shows two redox events as expected,[26] with half-wave potential ($E_{1/2}$) values of 0.73 (P_{a1}/P_{c1}) (P_a = anodic peak potential, P_c = cathodic peak potential) and −1.59 V (P_{a2}/P_{c2}) versus Ag/Ag$^+$, corresponding to the oxidation and reduction of PM567, respectively. As shown in Figure 1, after twenty five cycles the CV is largely unchanged save for a small response near 0 V versus Ag/Ag$^+$. The lack of significant changes indicates that, across the time scale of a CV experiment, PM567 shows good reversibility and electrochemical stability.

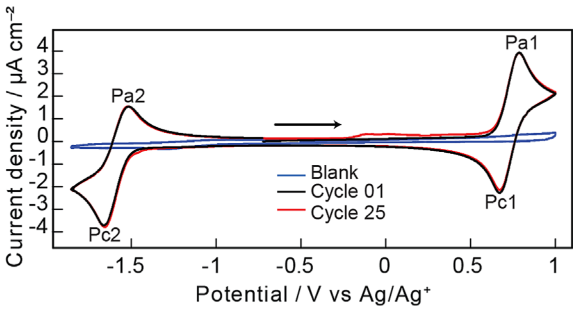

Figure 1. Cyclic voltammograms recorded with a scan rate of 0.5 V s^{-1} at a platinum working electrode in 6 mM PM567 and 100 mM $TBAPF_6$ in acetonitrile at room temperature.

Bulk electrolysis followed by cyclic voltammetry

The stability and electrochemical behavior of the oxidized/reduced states of a molecule are key determinants of efficiency as an active species in an energy storage device. The charged states of the active species must be stable with respect to chemical decomposition, membrane crossover, and self-discharge: all pathways that potentially reduce the efficiency of a device. A practical battery must remain charged for times far exceeding that of a typical CV experiment. With this in mind, chronoamperometry studies were carried out to obtain bulk quantities of charged species for stability studies. Both CV and UV/Vis spectroscopy were performed on the resulting oxidized and reduced materials.

Stability study of ferrocenium and cobaltocene

To validate these methods of assessing long-term stability, analogous studies were performed on ferrocene and cobaltocene, selected because of their celebrated reversible redox chemistry. Ferrocene and cobaltocene are electrochemically stable organometallic compounds and are often used as internal standards in redox reactions owing to their inherent inertness. Hwang et al. have exploited the promising electrochemistry of ferrocene and cobaltocene in redox flow cells.[28] They achieved a coulombic efficiency of approximately 78.6% across three cycles and a discharge potential of 1.1–1.4 V was obtained. During these pioneering studies, the ferrocene and cobaltocene were not stored in their charged states for any significant duration, prompting our study of the CV and UV/Vis absorbance of ferrocenium and cobaltocene over one month, paralleling our experiments on PM567.

Ferrocene was bulk oxidized to obtain ferrocenium in an H-cell (Figure S1). CVs were obtained before and immediately following bulk oxidation. As shown in Figure 2, the CV profiles of the ferrocene and ferrocenium do not significantly differ, with the only change being the value of the open circuit potential. No additional peaks appear, indicating that ferrocenium is stable far beyond the timescale of a typical CV experiment. UV/Vis spectroscopy data (Figure 3) showed no change over the course of 1 month, which further shows that ferrocenium is stable when stored under inert conditions.[29] Accordingly, RFBs using ferrocene as an active species will maintain charge provided that oxygen is excluded from the electrolyte solution. Similar experiments were carried out with cobaltocene. As expected, the CV profile of cobaltocene similarly remains stable upon reduction. UV/Vis studies show that the metallocene is stable for at least 1 month when stored under inert conditions (Figure S2 and Figure S3).

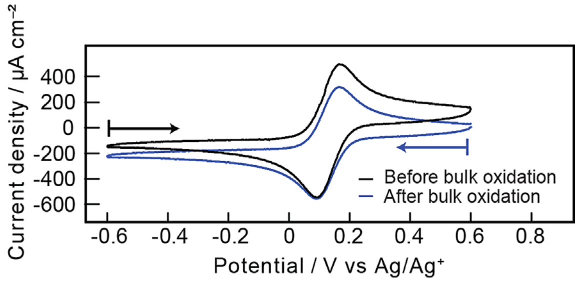

Figure 2. Cyclic voltammograms recorded with a scan rate of $0.5\ V\,s^{-1}$ at a platinum working electrode before and after the bulk oxidation of a solution of ferrocene and $TBAPF_6$ in acetonitrile at room temperature.

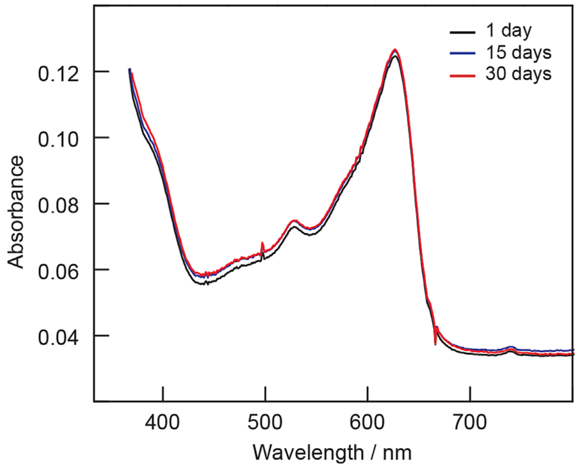

Figure 3. UV/Vis absorbance spectra of the bulk oxidized solution of ferrocene and $TBAPF_6$ in acetonitrile at room temperature.

Bulk oxidation of PM567

The CV of PM567 in acetonitrile indicates an $E_{1/2}$ of 0.73 V versus Ag/Ag^+. Accordingly, a potential of 1.3 V versus Ag/Ag^+ was applied to a platinum working electrode until the dye was fully oxidized. CVs of the resulting solution were taken at time points ranging from hours to days. Electronic absorption spectra were also obtained at each time point to provide further insight regarding any changes in composition. Figure S4 shows the current versus time curve for the bulk oxidation of PM567. The observation that the anodic current never reaches zero is consistent with previous experiments using PM567 reported by Sartin et al.[27] The persistent current was rationalized by diffusion of material from the counter electrode compartment through the separator that reduces a small amount of oxidized dye. The ongoing crossover of this reducing contaminant provides continuous substrate for oxidation and thus a non-zero current.

The CVs of the oxidized species taken at various time points are shown in Figure 4. Immediately after the bulk oxidation, the cyclic voltammetry profile of PM567 has changed. Two quasi-reversible redox couples occur at the same peak potentials as with neutral PM567. In addition, three anodic and two

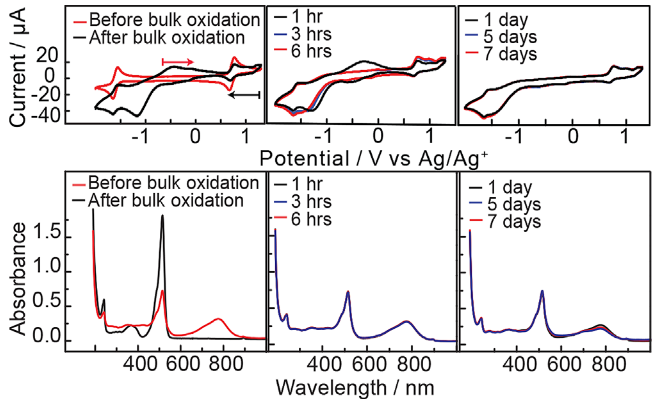

Figure 4. Cyclic voltammograms recorded at a platinum working electrode ($0.5\ V\,s^{-1}$ scan rate) and UV/Vis spectra of in the bulk oxidized solution PM567 and $TBAPF_6$ in acetonitrile at room temperature.

cathodic peaks appear, which are irreversible. This suggests that the radical ion species formed upon oxidation of PM567 is not stable over the 30 min required for bulk oxidation and setup of the CV experiment. Furthermore, the CVs continue to change over the course of 6 h, with the peak current density of the P_{a2}/P_{c2} couple decreasing over time. Despite the initial evolution of the voltammograms away from the idealized redox waves observed for the neutral dye, after 1 day a remarkable stability occurs, with no further change through a week of monitoring the CV responses. This lack of further change suggests that, although PM567 may initially degrade, the resulting solution still maintains a redox fidelity that allows for its use as an active species and motivated charge–discharge studies (vide infra). UV/Vis absorbance data further confirms that initial changes in composition stabilize after 24 h (Figure 4). A comparison between the UV/Vis of PM567 and its oxidized counterpart show striking changes immediately after electrolysis. Most notably a new band is observed at 770 nm and the intensity of the characteristic 514 nm band of BODIPY decreases by roughly half. This decrease in intensity is consistent with the loss of an electron that can no longer participate in the $\pi\rightarrow\pi^*$ transition. Although a small decrease in the band at 770 nm does occur over the course of 1 week, this minor change does not appreciably affect the redox response.

Bulk reduction of PM567

Bulk reduction was carried out with a working electrode potential of -2 V versus Ag/Ag^+. Similar to the bulk oxidation experiment, the change in CV profile was studied for 1 week with complementary monitoring by UV/Vis. Figure S5 shows the current versus time curve for the bulk reduction of PM567. Unlike the bulk oxidation curve, the current versus time curve describing the bulk reduction fully decays to the background. As seen in Figure 5, the CV of the bulk reduced species introduces three irreversible oxidation peaks. Unlike with the oxidation where some changes were observed over the course of hours, the reduced dye is even more stable after the initial induction period, persisting from hours to 1 week. The UV/Vis

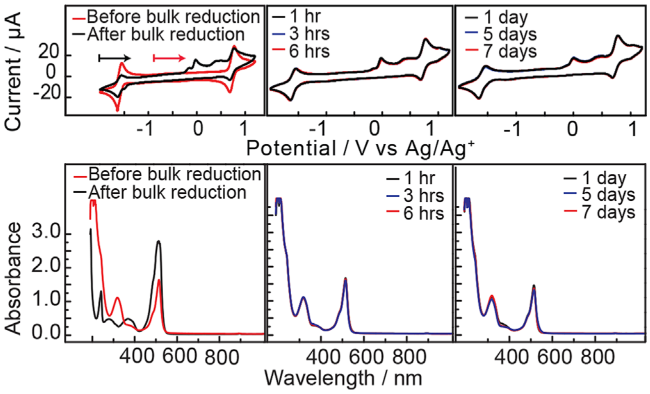

Figure 5. Cyclic voltammograms recorded at a platinum working electrode (0.5 V s^{-1} scan rate) and UV/Vis spectra of in the bulk oxidized solution PM567 and TBAPF$_6$ in acetonitrile at room temperature.

spectra of the bulk reduced species are shown in Figure 5. A similar attenuation in intensity of the 514 nm band occurs; however, no new low energy bands are introduced. As with the oxidation, slight changes occur over the course of days, but these small spectral differences do not have an impact on redox response.

Analyzing the products of bulk electrolysis

Initial investigations of chronoamperometry performed on BODIPY dyes invoke decomposition and potential polymerization of the radical species formed based on mass spectrometry results wherein peaks were observed at twice the *m/z* value of the parent BODIPY dyes.[27] The changes observed through CV prompted further mass spectroscopy analysis of the effects of oxidation and reduction products to determine the fate of the dye molecule (Figure 6). As shown in Figure S6, the mass spectrum of PM567 includes four main peaks, including one that corresponds to a dimer at an *m/z* value of 659. The isotopic distribution of these peaks is diagnostic of the presence of boron where a central peak dominates a given cluster.

In the oxidized PM567 spectrogram, three main peaks are observed. These are assigned to the species shown in Figure 7. The peak at an *m/z* of 571 corresponds to a dimeric species where one methyl group and one fluorine atom have fragmented from the dye. It should be noted that the peaks at *m/z* = 242 and 629 do not display characteristic isotopic distributions consistent with the presence of boron. The peak at 629 coincidentally matches the *m/z* value expected for a simple BODIPY dimer, a potential decomposition product that was suggested in a previous report.[27] However, the isotopic resolution illustrated in Figure 7 provides strong evidence that no boron remains in the species giving rise to this signal. A second entity also consistent with this *m/z* value and lacking BF_2 fragments is a dimer of the parent dipyrromethene with an associated PF_6^- from the supporting electrolyte.

In the Fourier-transform ion-cyclotron resonance mass spectrometer (FT-ICR-MS) of reduced PM567 (Figure 6), there are three main peaks. This mass spectrogram includes the peaks at

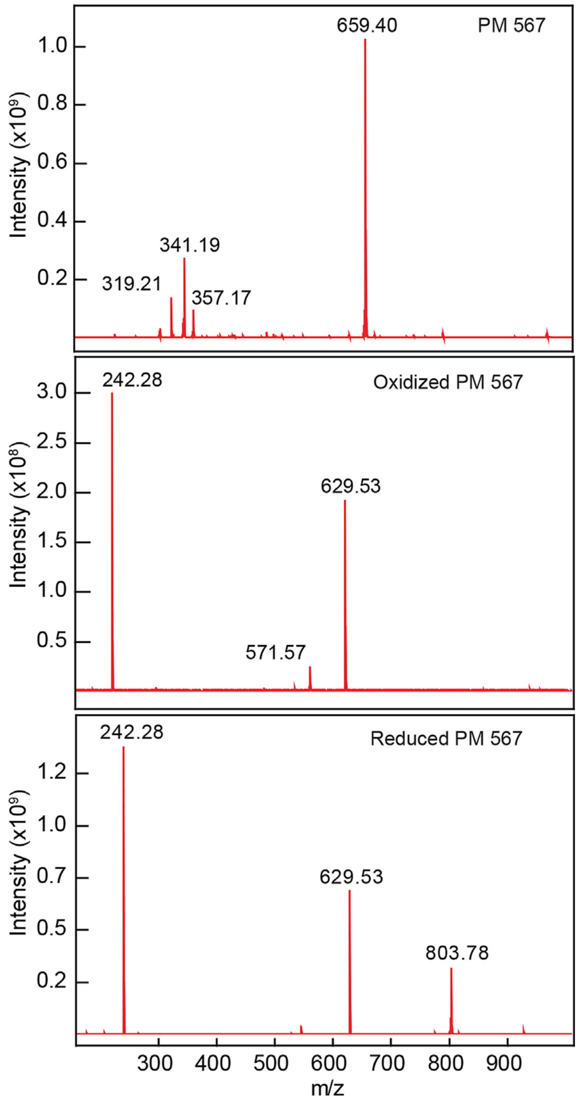

Figure 6. Experimental FT-ICR-MS of PM567, oxidized PM567, and reduced PM567.

m/z = 242 and 629 that were observed in the oxidation and result from the loss of the BF_2 moiety. As shown in Figure 7 the unique new peak that appears at *m/z* = 803 agrees well with a dimeric species that has lost two methyl groups and one fluorine from the PM567 core and includes one PF_6^-.

PM567 as a precursor for active species in an all-organic RFB

As observed from the CVs of the bulk electrolyzed PM567, the neutral PM567 is no longer the active species in the RFB. The new solution formed upon charging is a mixture of compounds partially identified by mass spectrometry. The final species responsible for redox activity show current response at similar potentials as neutral PM567 (Figure 4 and 5). Because the bulk species do not exhibit significant potential shifts, the theoretical cell voltage remains approximately 2.32 V once the system has stabilized.

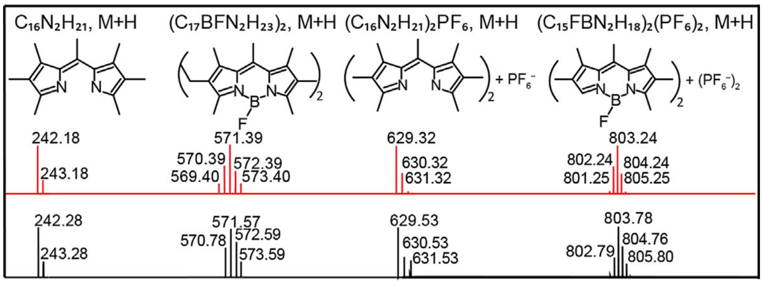

Figure 7. Calculated (red) and experimental (black) FT-ICR-MS peaks of oxidized and reduced PM567.

Diffusion coefficients and heterogeneous electron-transfer rate constants

An established method to determine the diffusion coefficients associated with a redox species uses a plot of peak current versus the square root of scan rate associated with a given redox event.[19–21] The Randles–Sevcik equation has been used to determine the diffusion coefficients from CV data obtained at platinum and glassy carbon working electrodes (ca. 10^{-5} cm^2 s^{-1}; Figure S7–S18). The Nicholson method was used to determine the heterogeneous electron-transfer rate constants at approximately10^{-2} cm s^{-1}.[30–32] These values are summarized in Table S1. As discussed above, PM567 acts as a precursor to a mixture of active species that form the oxidized and reduced dye. Because these are not isolable species, similar experiments to determine their diffusion coefficients and electron-transfer rate constants cannot be performed.

Solubility

The solubility of PM567 in acetonitrile was determined using electronic absorption spectroscopy of saturated solutions of the dye. Filtered stock solutions were diluted to absorbance intensities of less than 1.0. The molar absorptivity coefficient for PM567 was determined to be 7.9×10^4 M^{-1} cm^{-1}, in agreement with an earlier report.[33] Using the Beer–Lambert relationship, the maximum solubility of PM567 was 0.2 M in acetonitrile, taken as the average of three solutions. The least soluble component of an electrolyte couple in an RFB determines the upper limit of energy density. When the oxidized and reduced electrolyte materials are equally or more soluble than the neutral species, they will remain in solution upon redox cycling. We observe no precipitation in our PM567 system, indicating that the solubility of the neutral dye may be used to define an expected energy density. Using the calculated solubility of 0.2 M and a theoretical open circuit voltage of 2.32 V, energy density of the RFB can be estimated using the following equation [Eq. (1)]:

$$\text{Energy density} = 0.5\, nFV_{\text{cell}}c_{\text{active}} \quad (1)$$

where n is the number of electrons transferred, F is the Faraday constant, V_{cell} is the theoretical cell voltage, and c_{active} is the concentration of the redox species.

According to the Equation (1), an energy density of 22 kJ L^{-1} is estimated for a PM567 RFB. The energy density of all-vanadium RFBs reaches 90–126 kJ L^{-1} in some systems owing to the higher solubility of the active species.[34] These calculations use theoretical energy densities and will be affected when discharge voltages are reduced owing to cell design (e.g., high internal resistance and membrane crossover).

Charge–discharge experiments

A static cell was fabricated for initial performance evaluations of PM567 as an active species. This Teflon cell mimics the working compartments of an RFB system without any flow, enabling the use of small volumes. Initially, both compartments of the cell contained PM567 in a neutral state. Galvanostatic conditions were used with potential cut offs for both charge and discharge. A charging current of 0.12 mA and discharging current of 0.04 mA was used to ensure complete redox cycling (initial state-of-charge of 30%). A charge cut off voltage of 2.4 V was selected so as to slightly exceed the ideal 2.32 V cell potential predicted by CV experiments. The discharge cut off was set to 0.2 V. Figure 8 shows cycles 17–19 as representative curves of the cycling study. A charging plateau occurred around approximately 2.2 V, falling to 1.6 V upon discharge. Potential losses during discharge have been previously attributed mass-transport limitations[35] and high internal resistances associated with cell designs.[20] The resistance of the cell used here was measured across the two working electrodes prior to the first charge and after every five cycles. The resistance grew from 15 to 807 kΩ over the course of the charge–discharge experiment (Figure S20). The low conductivity of the electrolyte and the presence of a separator were two factors

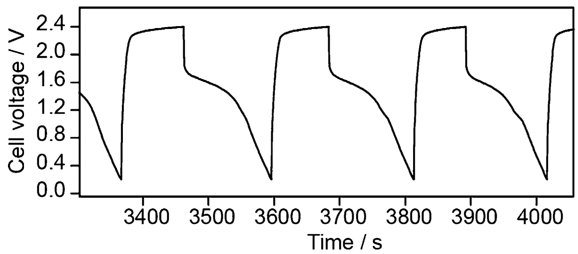

Figure 8. Representative constant current cycling curves for 1 mM PM567 and 100 mM $TBAPF_6$ in acetonitrile; charging current 0.12 mA and discharging current 0.04 mA; at room temperature.

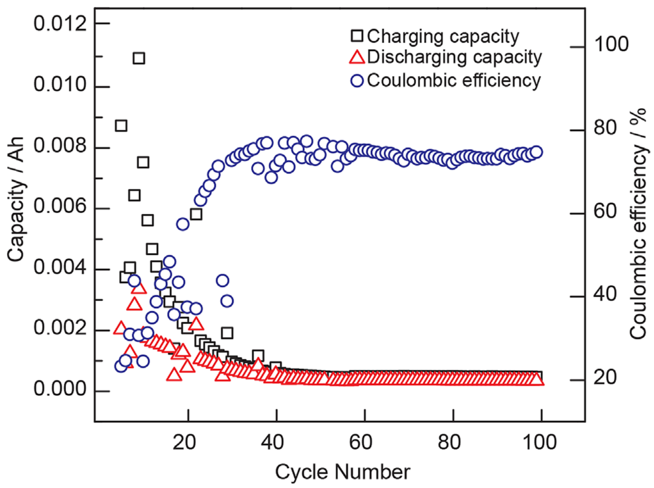

Figure 9. Charge capacity, discharge capacity, and coulombic efficiency of the PM567 cell.

contributing to the ohmic drop. As a result of these losses, the discharge voltages are lower than the cell voltages predicted by CV.

A long term cycling study was performed using 100 charge–discharge cycles. As shown in Figure 9 the charging capacity decreases throughout the first 40 cycles, consistent with dynamic capacities previously observed for a polymeric BODIPY study.[35] After an induction phase of 40 cycles, stable battery cycling was observed. The coulombic efficiency stabilized around 73%. Unknown side reactions, crossover of active species through the separator, and mass-transport limitations have all been invoked as factors governing the loss of coulombic efficiency in RFBs.[19,20] Although PM567 undergoes initial side reactions as evidenced by mass spectrometry and UV/Vis analyses of bulk electrolysis samples, both the oxidized and reduced solutions clearly stabilize, consistent with the plateau in coulombic efficiency observed during charge–discharge experiments.

Conclusions

A non-aqueous, all-organic redox flow battery (RFB) model system using a boron-dipyrromethene (BODIPY) dye (PM567) in tetrabutylammonium hexafluorophosphate ($TBAPF_6$)/acetonitrile was demonstrated using a laboratory-scale static-cell set up. The stability during storage of the oxidized/reduced species were studied using bulk electrolysis and UV/Vis spectroscopy. The radical cations/anions of PM567 were stable at the short timescale of cyclic voltammetry (CV), but this stability did not extend to longer time scales. Both UV/Vis and CV experiments showed significant changes over the course of hours that were in stark contrast to parallel experiments using ferrocene and cobaltocene, which remained quite stable in all charged states during storage. Despite the loss of well-defined oxidation and reduction waves for PM567, solutions of the dye did show reproducible redox behavior over the course of days to a week, indicating a promising stability for the purposes of charge–discharge cycles. The CV of these redox stabilized species predict an open circuit voltage of 2.32 V for a RFB with PM567 as the precursor for the active species. Mass spectrometry provided evidence for the loss of BF_2 fragments in both the oxidized and reduced states of the dye as a major decomposition product. The cycling performance of the cell with PM567 initially achieved a coulombic efficiency of 73% after a short induction period, making the dye an interesting example of a precursor that proves unstable upon charging, yet provides access to a redox stable mixture at longer timescales. These results suggest that stability and redox studies at short, medium, and long time-scales are all critical to fully assess potential active species. Molecules with promising redox couples based on CV may decompose, but this decomposition does not rule out use in RFB designs, as the resulting mixtures may show stable charge–discharge cycles.

Experimental Section

Electrolytes

Electrolyte solutions were prepared by dissolving the active species and $TBAPF_6$ (Alfa Aesar, USA) in anhydrous acetonitrile obtained from Grubbs-type solvent purification system (Pure Process Technology, USA). $TBAPF_6$ was recrystallized thrice using hot methanol and dried under vacuum. The active species PM567 (Sigma–Aldrich, USA), ferrocene (Sigma–Aldrich, USA), and cobaltocene (BTC, UK) were used as received.

Cyclic voltammetry experiments

Concentrations of PM567 and $TBAPF_6$ used were 6 mM and 100 mM, respectively. CV measurements were carried out using a Bio-Logic SP 300 potentiostat/galvanostat and the EC-Lab software suite. Both 3 mm diameter glassy carbon discs and 2 mm diameter platinum discs (CH Instruments, USA) were used as working electrodes. The working electrodes were polished with a micro cloth pad using 0.05 μM alumina powder (CH Instruments, USA). After polishing, the electrodes were ultrasonically cleaned and rinsed thoroughly with acetone and deionized water, and allowed to dry before use. Potentials during CV were measured relative to a non-aqueous Ag/Ag^+ reference electrode with 10 mM $AgNO_3$ and 100 mM $TBAPF_6$ in acetonitrile solvent (BASi, USA). A platinum wire (CH Instruments) served as the counter electrode. All experiments were carried out at room temperature inside a nitrogen-filled glove box (Vigor Tech, USA). All CV measurements were iR compensated at 85% with impendence taken at 100 kHz using the ZIR tool included within the EC-Lab software.

Bulk electrolysis experiments

Bulk electrolysis experiments were performed in a two compartment H-cell (Adams and Chittenden, USA) inside a nitrogen-filled glove box. The compartments of the H-cell were separated by a glass frit (porosity 10–16 μm). The potential at the working electrode was controlled with reference to an Ag/Ag^+ non aqueous reference electrode. A platinum mesh was employed as the working electrode and a platinum wire served as the counter electrode. The reference electrode and the working electrode occupied the same compartment containing 20 mL of the solution of 6 mM PM567 and 100 mM supporting electrolyte in acetonitrile. The counter electrode compartment contained 20 mL of 100 mM

$TBAPF_6$ in acetonitrile solution. After bulk electrolysis, the solution in the working electrode compartment was transferred to a three electrode cell for CV measurements.

Solubility determination

A saturated solution was prepared by the sequential addition of solid PM567 into acetonitrile (5 mL) with stirring until a suspension was formed. The solution was allowed to settle for 2 h and subsequently filtered through glass wool to remove any undissolved material. An aliquot of this solution was diluted in acetonitrile and the UV/Vis spectrum was recorded on a Cary 8454 UV/Vis diode array system. The absorbance recorded at 514 nm for three independent trials was used to determine the maximum solubility of PM567 in acetonitrile. The molar absorption coefficient of PM567 was determined using three stock solutions serially diluted to absorbances between 1.0 and 0.1 using the Beer–Lambert relationship.

UV/Vis spectroscopy

UV/Vis absorbance of the electrolyte solutions before and after bulk electrolysis were measured using a Cary 8454 UV/Vis spectrometer and a 1 cm path length airtight cuvette (Starna, USA).

Mass spectrometry

Mass spectra were collected using a Bruker Daltonics SolariX 12 tesla FT-ICR-MS.

Charge–discharge experiments

A custom-made Teflon cell was used for charge–discharge experiments (Figure S19). Each compartment contained 1 mL of the 1 mM of the active species in 100 mM $TBAPF_6$. The electrolytes were separated by an AMI-7001 anion exchange membrane (2 cm × 1 cm, Membrane International Inc., USA). Prior to each experiment, the membrane was pre-conditioned by soaking in a solution containing 100 mM $TBAPF_6$ in acetonitrile for 24 h. Two graphite felt electrodes (0.9 × 0.6 × 1 cm, Fuel Cell Store, USA) were used. The electrode was submerged (ca. 0.7 cm in the solution) leaving a small volume at the top to connect platinum-wire current collectors to which leads were attached. The electrodes were placed with a small gap from the membrane. Galvanostatic conditions were employed with charge and discharge currents of 0.12 mA and 0.04 mA for the PM567 system and 0.05 mA and 0.01 mA for the ferrocene/cobaltacenium hexafluorophosphate system, respectively. The cells were charged up to overall cell potentials of 2.4 V (PM567) and 1.38 V (ferrocene/cobaltacenium hexafluorophosphate) and discharged to 0.2 V. All charge–discharge experiments were performed in a nitrogen-filled glove box.

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Keywords: BODIPY · cyclic voltammetry · electrolyte · redox flow battery · stability

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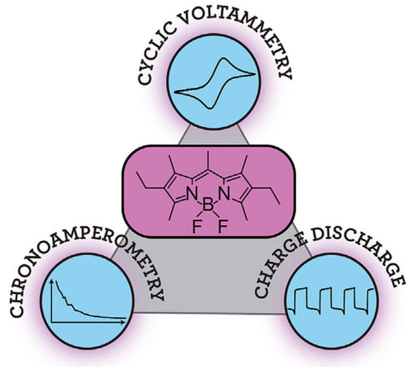
Characterization of a BODIPY Dye as an Active Species for Redox Flow Batteries



Bits & pieces: A redox flow battery employing a fluorescent boron-dipyrromethene (BODIPY) dye (PM 567) is investigated. A theoretical cell potential of 2.32 V is predicted from cyclic voltammetry (CV) experiments. To evaluate stability, bulk electrolysis and CV experiments together with UV/Vis absorbance spectroscopy are performed. Oxidized and reduced, PM 567 does not remain fully intact. However, the products of bulk electrolysis evolve over time to show stable redox behavior suitable for redox flow batteries.